

Accuracy of discriminant analysis methods to classify activities of subjects with Parkinson's Disease using wearable sensors

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Abstract

Research Objectives

To investigate the accuracy of four discriminant analysis methods to classify activities of subjects with Parkinson's Disease (PD) to support remote monitoring of performance using wearable sensors.

Design

Activities were classified using computational methods and compared to manually tagged activities performed in random order by PD subjects during a cross-sectional study.

Setting

Activities were performed in the Motion Analysis lab at an outpatient rehabilitation center.

Participants

Five PD subjects (mean age: 63.5 $\pm$ 2.6 years, mean Hoehn-Yahr score: 2) performed activities during the on-medication state.

Interventions

Four wearable sensors were used to capture the motion of subjects during a parkour of six activities. Motion data were used to train and test four discriminant analysis methods.

Main Outcome Measures

Three-dimensional components and magnitudes of acceleration and angular rate of sensors located on the foot, thigh, pelvis, and wrist were registered at a sampling rate of 250 Hz. Data were preprocessed using a lowpass 2nd order Butterworth filter with a cutoff frequency of 5 Hz and smoothed using a moving-average filter with a window size of 6.4 s. Thirteen predictors were used to train and test 4 discriminant analysis methods: linear, quadratic, naive Bayes (NB) with normal, and NB with kernel distribution. The skewness and normality of each predictor were calculated. Discriminant analysis methods were validated using the 5-fold cross-validation method. Accuracy for each method was calculated. All data processing analysis was performed in Matlab 2020.

Results

A total of 179,555 observations were used to train and validate discriminant analysis methods. None of the predictors showed a normal distribution and 6 of them presented high skewness. The lowest accuracy was obtained by linear discriminant analysis (55.9%). Best accuracy was obtained for naive Bayes with kernel distribution (60.7%) because it does not require normally distributed data and supports skewness in predictors.

Conclusions

Discriminant analysis using naive Bayes with kernel distribution has the potential to identify activities that could be helpful to support remote monitoring of the performance of PD subjects using wearable sensors.

Key Words

Accuracy, Machine Learning, Artificial Intelligence, Parkinson's Disease, Wearable Sensors